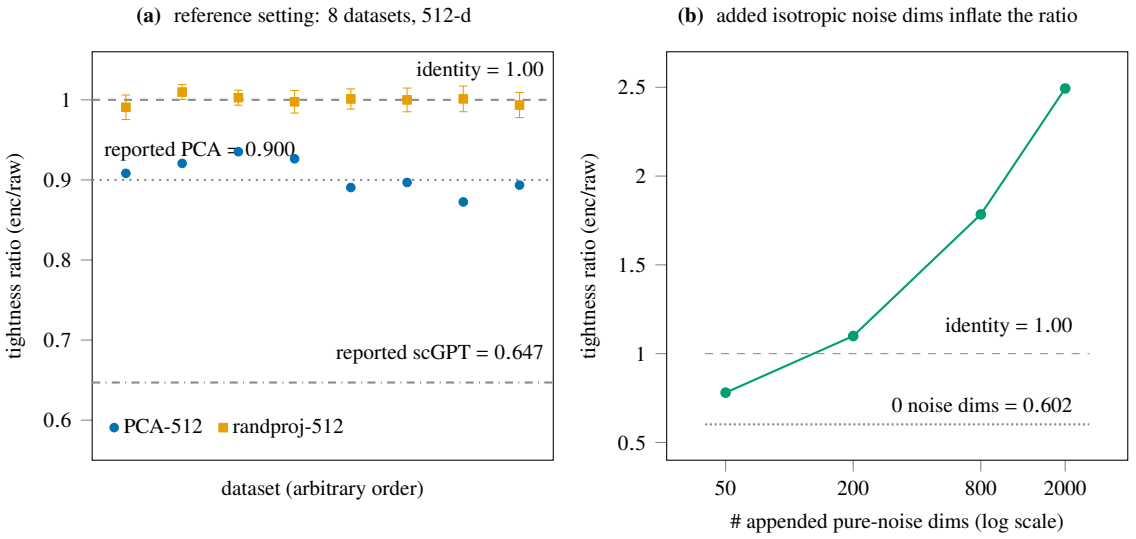

(a) reference setting: 8 datasets, 512-d
identity = 1.00
reported PCA = 0.900
reported scGPT = 0.647
PCA-512
randproj-512
tightness ratio (enc/raw)
dataset (arbitrary order)
1
0.9
0.8
0.7
0.6
(b) added isotropic noise dims inflate the ratio
identity = 1.00
0 noise dims = 0.602
tightness ratio (enc/raw)
appended pure-noise dims (log scale)
2.5
2
1.5
1
0.5
50
200
800
2000